



Customer Shopping Behavior Analysis

Uncovering insights from transactional data to guide strategic decisions.

Project Overview

Data Analysis

Analyzed 3,900 purchases across categories.

Key Insights

Spending patterns, segments, product preferences, subscriptions.

Strategic Guidance

Informing business decisions with data-driven insights.



Dataset Summary

Rows

3,900

Columns

18

Key Features include customer demographics, purchase details, and shopping behavior. Missing data in Review Rating column (37 values).

Exploratory Data Analysis (Python)

01

Data Loading & Exploration

Pandas for import, df.info() and .describe() for summary.

02

Missing Data Handling

Imputed Review Rating with median by category.

03

Column Standardization

Renamed columns to snake_case for readability.

04

Feature Engineering

Created age_group and purchase_frequency_days.

05

Data Consistency

Dropped redundant promo_code_used column.

06

Database Integration

Loaded cleaned data to PostgreSQL for SQL analysis.

Report

Data Analysis (SQL)

Revenue by Gender

Female: \$75,191

Male: \$157,890

Top 5 Products by Rating

Gloves: 3.86

Sandals: 3.84

Boots: 3.82

Hat: 3.80

Skirt: 3.78

Shipping Type Comparison

Standard: \$58.46

Express: \$60.48

Subscribers vs. Non-Subscribers

Subscribers: 1,053 customers, Avg Spend \$59.49, Total Revenue \$62,645

Non-Subscribers: 2,847 customers, Avg Spend \$59.87,
Total Revenue \$170,436

SQL Insights: Customer Behavior

1

High-Spending Discount Users

839 customers spent above average with discounts.

2

Discount-Dependent Products

Hat (50%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), Pants (47.37%).

3

Customer Segmentation

Loyal (3116), Returning (701), New (83).

4

Repeat Buyers & Subscriptions

958 repeat buyers are subscribers, 2518 are not.

SQL Insights: Product & Revenue

Top 3 Products per Category

Accessories: Jewelry, Sunglasses, Belt

Clothing: Blouse, Pants, Shirt

Footwear: Sandals, Shoes, Sneakers

Outerwear: Jacket, Coat

Revenue by Age Group

Young Adult: \$62,143

Middle-aged: \$59,197

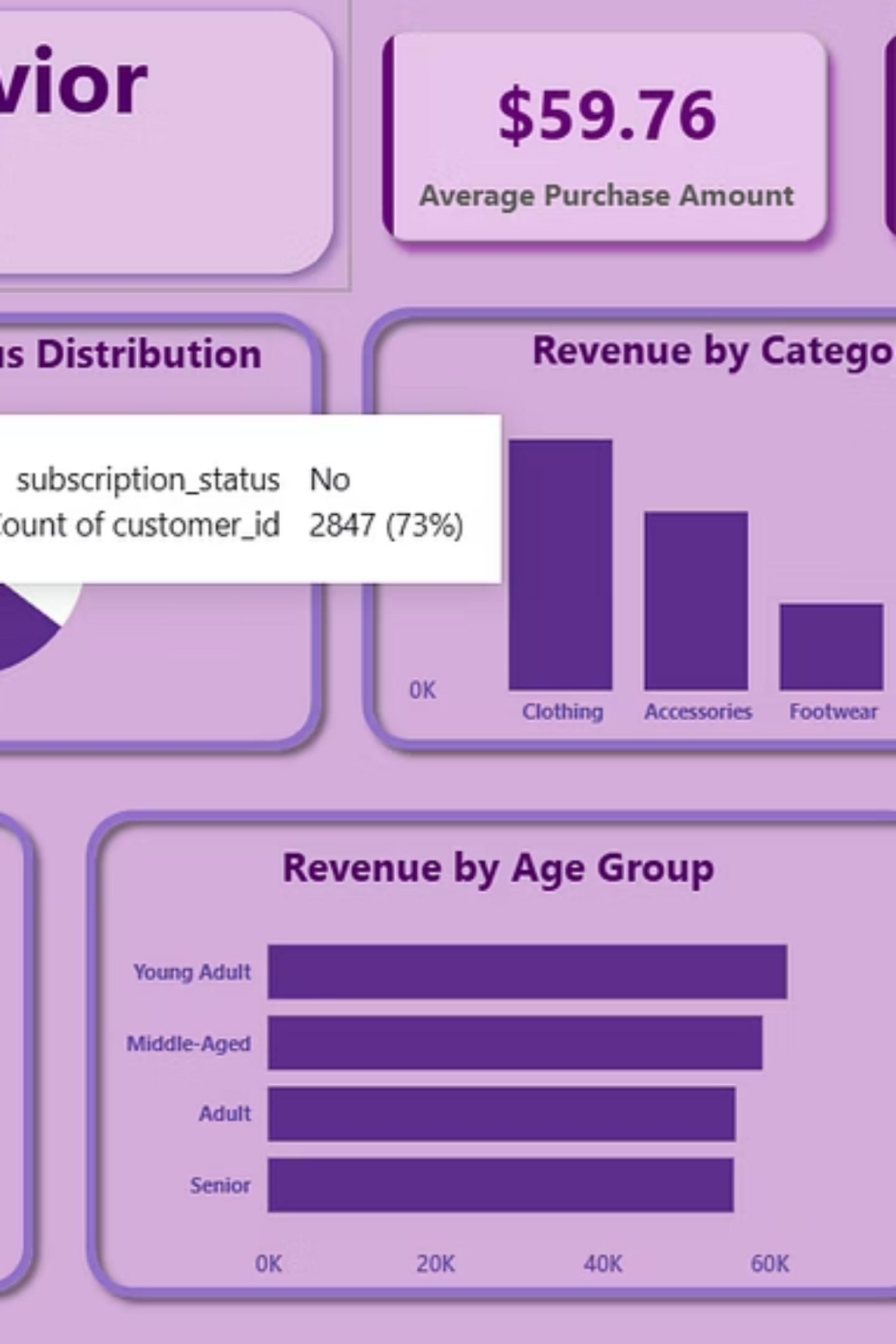
Adult: \$55,978

Senior: \$55,763



Power BI Dashboard

Interactive visualization of customer behavior insights.



Dashboard Key Metrics

3.9K

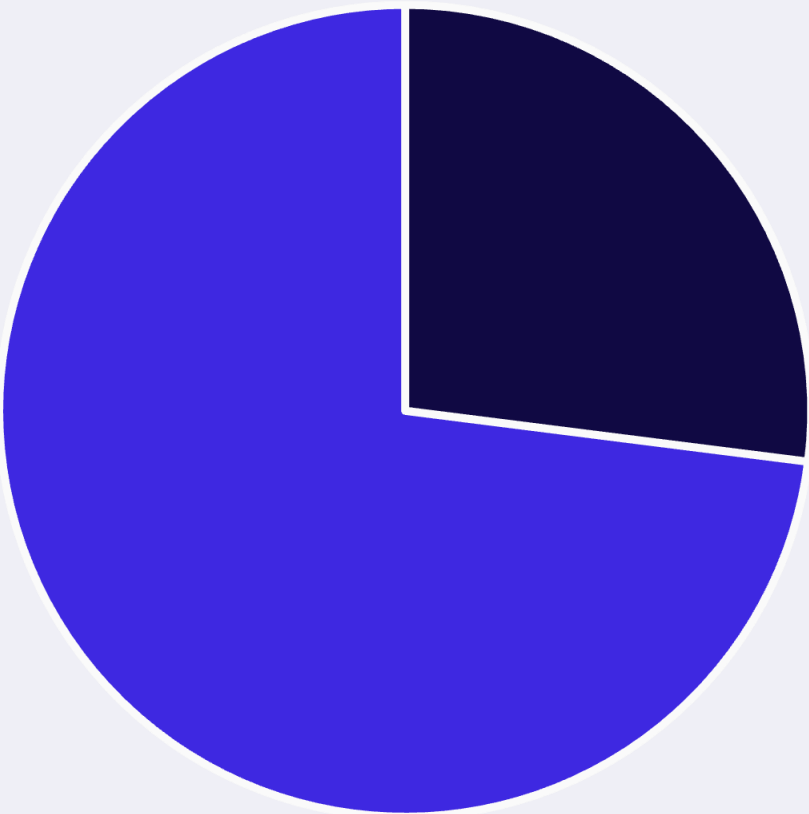
Customers

\$59.76

Avg. Purchase

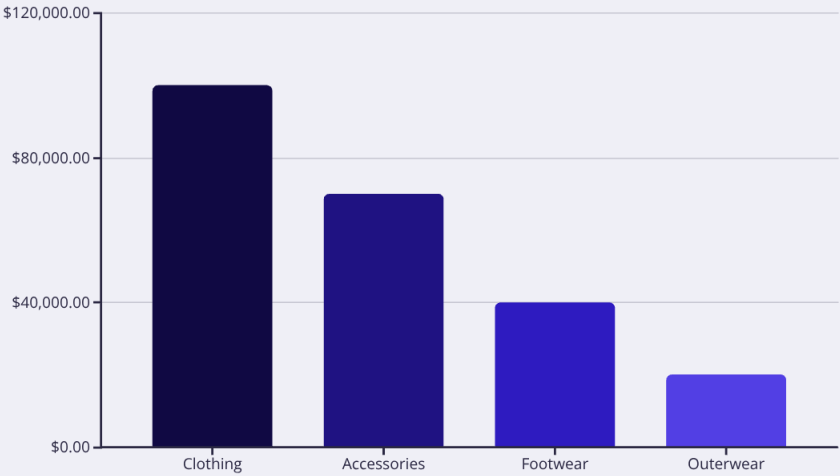
3.75

Avg. Review



Yes No

27% of customers are subscribers.



Clothing drives the most revenue.

Business Recommendations

→ Boost Subscriptions

Promote exclusive benefits for subscribers.

→ Customer Loyalty Programs

Reward repeat buyers to foster loyalty.

→ Review Discount Policy

Balance sales with margin control.

→ Product Positioning

Highlight top-rated and best-selling products.

→ Targeted Marketing

Focus on high-revenue age groups and express-shipping users.